LarvaLens

Product Requirements Document - PRD

The product north star: problem, users, value proposition, MVP features, exclusions, user stories, success metrics and safe product language.

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Vibe Document 01 - Product Requirements Document

Document ID: `01_PRD.md` Status: Source of truth Product: LarvaLens Tagline: Video-based probable mosquito-larva screening with debris rejection and geotagged evidence.

6.1 Problem

Field workers and communities often inspect stagnant water manually. Static-image classifiers can confuse reflections, sand, leaves, bubbles and floating particles with larvae, and a single detection confidence does not explain why an object was accepted. LarvaLens uses short video, candidate detection, binary larva-vs-non-larva verification and temporal tracking to create reviewable evidence.

LarvaLens is a screening and surveillance prototype. It is not a medical diagnosis, definitive species identification system or replacement for entomological confirmation.

6.2 Primary users

Persona	Need
Field surveyor / sanitation worker	Capture or upload a 5-10 second water video, receive an explainable probable-larva result and geotag the observation.
Public-health reviewer	Inspect accepted and rejected tracks, confirm or reject a scan and view hotspot patterns.
Project administrator	Verify model versions and service health without viewing user secrets.
Public viewer	View aggregated hotspot information without reporter identity or raw private video.

6.3 Core value proposition

LarvaLens does not trust a single still-frame prediction. It combines where an object appears, whether its crop resembles larval morphology rather than debris, and whether it persists/moves across time. Every result records the model versions and track-level acceptance/rejection reason.

6.4 Must-have features for the MVP

- Email/password authentication through Supabase.
- Responsive camera/upload flow for short video.
- Optional browser geolocation with explicit permission and manual skip.
- Server-side file validation, private storage upload and real scan row.
- Required model-readiness gate before accepting analysis.
- Detector + binary verifier + temporal tracking integration.
- Real status/progress updates from the database.
- Result with probable-larva count, rejected-track count, confidence, quality and risk band.
- Track evidence showing accepted/rejected state and reason.
- Scan history for the signed-in user.
- Geotagged hotspot map using aggregated scan results.
- Reviewer queue and human decision for authorized users.
- No production mock data, hard-coded predictions or external vision AI.

6.5 Nice-to-have features

- Offline capture queue in IndexedDB, uploaded when connectivity returns.
- Annotated output-video generation.
- Experimental species hint only for high-detail crops.
- Exportable CSV/PDF field report.
- Multilingual UI labels.
- Weather overlay and ward-level analytics after the core flow is stable.

6.6 Explicitly out of scope

- Training models inside the web/backend repository.
- Autonomous public-health action without human confirmation.
- Guaranteed mosquito species identification from ordinary phone video.
- Live camera inference entirely inside the browser.
- Native Flutter/Kotlin application during the initial eight-hour build.
- Clinical/epidemiological claims or a fixed claimed accuracy before field testing.
- Public exposure of raw videos, precise home coordinates or user identity.

6.7 User stories

- As a field surveyor, I want to upload a short water video so I can receive a traceable probable-larva result.
- As a field surveyor, I want to see actual processing stages so I know the system has not frozen.
- As a field surveyor, I want a clear retake reason when the video is too dark, blurry or unstable.
- As a reviewer, I want accepted and rejected track evidence so I can audit false positives.
- As a reviewer, I want to override the final review decision without erasing the original model evidence.
- As an administrator, I want the active model hashes and readiness so I can detect a wrong checkpoint.
- As a public viewer, I want aggregated hotspot information without seeing personal data.

6.8 Success metrics

Metric	MVP success criterion
End-to-end completion	A real video produces a real stored result without manual database editing.
Data integrity	Every completed scan stores detector/verifier hashes and track evidence.
Negative-clip rejection	Report actual `TN/(TN+FP)` on a held-out field set; no invented target.
Positive-clip detection	Report actual `TP/(TP+FN)` on a held-out field set.
UX	User can understand capture, processing, result and retake states on a 360px phone screen.
Failure honesty	Missing models, invalid files, unavailable location and inference errors show explicit states; none become fake success.

6.9 Product language rules

Use "probable mosquito larvae detected," "screening result," "requires review," and "no probable larvae detected in this clip." Avoid "disease diagnosed," "water is safe," "100% accurate," or "mosquito species confirmed."